Fish sauce (fermented anchovy/marine sauce): HMTc Standard

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CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

Abstract

The Heavy Metal Tested & Certified standard for Fish sauce (fermented anchovy/marine sauce) (Condiments, Sauces and Dressings), stated as consumed. Each analyte carries the strictest applicable government maximum level in the product's native basis, with a disclosed read-across or a mandatory control where no government regulates the exact product.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default; face value, no reconstitution; read-across / control / not-material for unregulated cells; no occurrence percentiles)

2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

ANALYTE	HMTC LIMIT	BASIS
Lead (Pb)	300	human-grade read-across (European Union)
Cadmium (Cd)	50	human-grade read-across (CR)

Total arsenic (tAs)	2,000	human-grade read-across (Australia / New Zealand)
Inorganic arsenic (iAs)	100	human-grade read-across (European Union)
Mercury, total (tHg)	300	human-grade read-across (European Union)
Methylmercury (MeHg)	300	bounded by tHg (reflex speciation)
Nickel (Ni)	20	human-grade read-across (European Union)
Tin (Sn)	100,000	human-grade read-across (European Union)
Aluminium (Al)	200	human-grade read-across (European Union)
Total chromium (Cr)	50	human-grade read-across (AU)
Hexavalent chromium (Cr-VI)	≤ Cr	bounded by Cr (reflex speciation)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	300	European Union — human-food analog
Cadmium (Cd)	50	CR — human-food analog
Total arsenic (tAs)	2,000	Australia / New Zealand — human-food analog
Inorganic arsenic (iAs)	100	European Union — human-food analog
Mercury, total (tHg)	300	European Union — human-food analog

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Methylmercury (MeHg)	300	Methylmercury is a component of total mercury (MeHg <= tHg). The MeHg value (500 ug/kg) is not below the binding total mercury ceiling (300 ug/kg), so methylmercury is governed by and bounded by the total mercury ceiling; no separate MeHg ceiling is published. It is enforced by reflex speciation whenever tHg is elevated.
Nickel (Ni)	20	European Union — human-food analog
Tin (Sn)	100,000	European Union — human-food analog
Aluminium (Al)	200	European Union — human-food analog
Total chromium (Cr)	50	AU — human-food analog
Hexavalent chromium (Cr-VI)	≤ Cr	Cr-VI carries no product-specific sovereign cell; it is a component of and bounded by its parent total Cr (Cr-VI <= Cr), an enforceable ceiling via reflex speciation rather than an empty cell.

4 | Cite this standard

Pendergrass, K. (2026). *Fish sauce (fermented anchovy/marine sauce): HMTc standard* (Standard No. 10.83138/hmtc.std.32.fish-sauce). Institute of Contaminant Standards. https://doi.org/10.83138/hmtc.std.32.fish-sauce

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References

[1] AU — human-food maximum levels
Binds Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] Australia / New Zealand — human-food maximum levels
Binds tAs. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] CR — human-food maximum levels
Binds Cd. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[4] European Union — human-food maximum levels
Binds Pb, iAs, tHg, Ni, Sn, Al. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/fish-sauce. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/fish-sauce.